2024, Mar. 2024 Release LOC:Acute Adult
Heart Failure

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview

Introduction:
Heart failure (HF) is a clinical syndrome which occurs when the myocardium becomes unable to pump blood effectively and causes passive congestion of the pulmonary or venous vasculature. HF may start acutely or evolve into a chronic condition. Acute HF is a life-threatening condition and requires urgent treatment.
Major risk factors for developing HF include myocardial infarction (MI), coronary artery disease (CAD), hypertension, valvular heart disease, congenital heart defects, arrhythmias, viral infections, obesity, and exposure to cardiac toxins (e.g., alcohol, tobacco, cocaine). Additionally, certain medications can cause myocardial toxicity or exacerbate underlying myocardial dysfunction and precipitate acute HF (e.g., chemotherapy, nonsteroidal anti-inflammatory drugs (NSAIDs), antidepressants, antihypertensive medications). Acute exacerbation of chronic HF is often precipitated by non-adherence to medications or dietary restrictions (Heidenreich et al., Circulation 2022, 145: e895-e1032; Page et al., Circulation 2016).
The cardinal manifestations of HF are dyspnea upon exertion or at rest, fatigue, weakness, lower extremity edema, rapid or irregular heartbeat, decreased exercise tolerance, persistent cough or wheezing with production of pink, frothy sputum, nocturia, sudden weight gain, anorexia, and chest pain.
The New York Heart Association (NYHA) developed the following classification system:

- **Class I**: No symptoms with ordinary physical activity
- **Class II**: Mild symptoms and slight limitations with ordinary physical activity
- **Class III**: Marked symptoms and limitations even with light physical activity
- **Class IV**: Symptoms at rest with severe limitations

The American College of Cardiology/American Heart Association (ACC/AHA) developed a system to identify structural disease and risk factors that contribute to morbidity and mortality:

- **Stage A**: High risk for HF but without structural heart disease or symptoms
- **Stage B**: Structural heart disease, increased filling pressure, or otherwise unexplained elevated biomarkers (troponin or natriuretic peptide biomarkers) without symptoms of HF
- **Stage C**: Structural heart disease with prior or current symptoms
- **Stage D**: Refractory HF requiring specialized interventions

Patients with established diagnosis of HF may also be classified based on their left ventricular ejection fraction (LVEF), also referred as ejection fraction (EF) (Heidenreich et al., Circulation 2022, 145: e895-e1032):

- **HF with reduced EF (HFrEF)**: LVEF ≤ 40%
- **HF with improved EF (HFimpEF)**: LVEF was ≤ 40% previously but follow up measurement now shows LVEF > 40%
- **HF with mildly reduced EF (HFmrEF)**: LVEF 41-49%
- **HF with preserved EF (HFpEF)**: LVEF ≥ 50%

Patients with advanced heart failure (HF) may have a history of the following (Heidenreich et al., Circulation 2022, 145: e895-e1032; Yancy et al., J Card Fail 2017, 23: 628-51):

- Repeated hospitalizations or emergency department (ED) visits for HF in the past year
- Progressive deterioration in renal function with a rise in blood urea nitrogen (BUN) and creatinine
- Weight loss without other cause (e.g., cardiac cachexia)
- Intolerance to renin-angiotensin-aldosterone system inhibitors due to hypotension or worsening renal function
- Intolerance to beta blockers due to worsening HF or hypotension
- Frequent systolic blood pressures less than 90 mm Hg
- Continued NYHA functional class III or IV symptoms despite HF therapy or requiring intravenous inotropes
- Inability to walk one block on level ground due to dyspnea or fatigue
- Recent need to escalate diuretics to maintain volume status, often reaching daily furosemide equivalent dose greater than 160 mg per day or use of supplemental metolazone therapy
- Progressive decline in serum sodium, usually to less than 134 mEq/L
- Frequent shocks from implantable cardioverter defibrillator (ICD)

Evaluation and Treatment:
The term guideline-directed medical therapy (GDMT) has been introduced to include evaluation and treatments that may be involved in the care of patients with congestive HF. This may include clinical and laboratory assessment, diagnostic testing, procedures, and medications. There are 4 classes of medications recommended in GDMT:

- Angiotensin converting enzyme inhibitors (ACEi) or angiotensin receptor blockers (ARB) or angiotensin receptor neprilysin inhibitors (ARNi)
- Beta blockers
- Mineralocorticoid receptor antagonists (MRA)
- Sodium-glucose cotransporter 2 inhibitors (SGLT2i)

Medication used for each patient may differ based on severity of disease and relative contraindications. SGLT2i are a relatively new class of medication and have been found to be associated with reduced risk of hospitalization and cardiovascular mortality, whether or not there is pre-existing type 2 diabetes. Other medications that may also be utilized include diuretics, isosorbide and hydralazine, ivabradine, and digoxin (Heidenreich et al., Circulation 2022, 145: e895-e1032).
The initial evaluation of suspected exacerbation of HF seek to determine the severity of cardiopulmonary instability by assessing the degree of hypoxemia and the hemodynamic status. Standard diagnostic exams and testing may include:

- Physical examination: vital signs, mental status, weight gain, jugular venous distention, breath sounds, heart sounds, peripheral edema
- Laboratory studies: natriuretic peptide biomarker, troponin, blood urea nitrogen (BUN), creatinine, electrolytes, glucose, complete blood count
- Other studies: chest X-Ray (CXR), electrocardiogram (ECG), echocardiogram

B-type natriuretic peptide (BNP) and N-terminal pro-BNP (NT-proBNP) are serum markers released by the cardiomyocytes in response to ventricular stretch and strain. BNP levels may be elevated with acute HF decompensation. Other conditions that can cause elevated natriuretic peptide levels include acute coronary syndrome (ACS), hypertension with left ventricular hypertrophy (LVH), valvular heart disease, atrial fibrillation, pulmonary embolism, pulmonary hypertension, sepsis, chronic obstructive pulmonary disease (COPD), renal dysfunction, or hyperthyroidism. Factors such as age, sex, weight, and renal function can influence BNP levels. BNP values should be used in conjunction with symptoms, physical examination, and radiological findings to aid in the diagnosis of HF.
Treatment of acute congestive HF is focused on hemodynamic stabilization, support of oxygenation or ventilation, and decongestion. For patients requiring hospitalization for acute decompensation, the following medications may be used: intravenous diuretics and vasodilators, GDMT medications, inotropes, and nitrates. Oxygen therapy should be considered for patients with an O_2 saturation less than 90%. Depending on the underlying etiology, other measures to treat refractory HF may include coronary revascularization (e.g., coronary artery bypass graft (CABG), percutaneous coronary intervention (PCI)), heart valve repair or replacement, implantable cardioverter-defibrillator (ICD) or

biventricular pacemaker, cardiac resynchronization therapy (CRT), ventricular assistive devices (VAD), and heart transplant (Heidenreich et al., Circulation 2022, 145: e895-e1032; McDonagh et al., Eur Heart J 2021, 42: 3599-726; Yancy et al., J Card Fail 2017, 23: 628-51; Mebazaa et al., Eur J Heart Fail 2015, 17: 544-58).
Many HF patients are at high risk for readmission, and careful attention to discharge planning is essential.

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources (Heidenreich et al., Circulation 2022, 145: e895-e1032; McDonagh et al., Eur Heart J 2021, 42: 3599-726; Hollenberg et al., J Am Coll Cardiol 2019, 74: 1966-2011; Yancy et al., J Card Fail 2017, 23: 628-51; Yancy et al., Circulation 2013, 128: e240-327).

(Excludes PO medications unless noted)

Select Day, One:

- **Initial review, One:** [1, 2, 3, 4, 5, 6]
  (From arrival in ED through decision to admit)
  (Symptom or finding within 24h)
  - **OBSERVATION,** ≥ **One:** [7]
    - Dyspnea, not returned to baseline after at least 1 dose of diuretic and ≥ 2h treatment and, ≥ **One:** [8, 9, 10]
      - O_2 sat 89-91%(0.89-0.91) and < baseline [9, 11]
      - Edema of lower extremities
      - Hepatomegaly
      - Jugular venous distention
      - Rales (crackles) on physical examination
      - Pleural effusion or pulmonary edema or cardiomegaly confirmed by CXR
      - Weight gain of ≥ 3 lbs (1.4 kg) over last 2d or ≥ 5 lbs (2.3 kg) in 7d [12]
    - Failed outpatient management, **Both:** [13]
      - Dyspnea or weight gain of ≥ 3 lbs (1.4 kg) over last 2d or ≥ 5 lbs (2.3 kg) in 7d [12]
      - ≥ 2 visits within last 7d
  - **ACUTE, One:** [14]
    - Heart failure, new onset and, **Both:** [15]
      - Symptom, ≥ **One:** [16]
        - Dyspnea
        - Orthopnea
        - Paroxysmal nocturnal dyspnea
      - Finding, ≥ **One:** [17]
        - Rales (crackles) on physical examination
        - Gallop (S3, S4) on physical examination
        - Pleural effusion or pulmonary edema or cardiomegaly confirmed by CXR
        - Edema of lower extremities
        - Hepatomegaly
        - Jugular venous distention
        - BNP or NT-proBNP > ULN [18]
    - Heart failure, acute on chronic and, **Both:**
      - Continued symptoms after at least 1 dose of diuretic and ≥ 2h treatment and, ≥ **One:** [10, 16]
        - Dyspnea, not returned to baseline [9]
        - O_2 sat < 89%(0.89) and < baseline [9, 11]
      - Finding, ≥ **One:**
        - Inadequate diuresis, ≥ **One:** [19]
          - Urine sodium < 50 mEq/L(50 mmol/L) at 2h after diuretic [20]
          - Urine output < 150 ml/h at 2h after diuretic [20]
        - Clinical risk factor, ≥ **One:** [21]
          - Troponin > ULN [22]
          - Creatinine ≥ 1.5x baseline or estimated baseline [23, 24]
          - Chronic kidney disease (excludes chronic dialysis) and creatinine ≥ 2.75 mg/dL [25]
          - BUN ≥ 43 mg/dL [26]
          - Sodium < 130 mEq/L(130 mmol/L) [27]
          - Heart rate 100-120/min, sustained [28, 29]
          - SBP < 120 mmHg and < baseline [30]
          - Frailty, **Both:** [31, 32]
            - Age ≥ 65 yrs
            - Finding, ≥ **One:**
              - Frailty, active diagnosis [33]
              - Frailty characteristics, ≥ **Two:** [34]
                - Chronic condition(s), ≥ **One:**

- Dementia or cognitive impairment [35]
- Multimorbidity, ≥ 2 chronic illnesses [36, 37]
- Functional impairment [38, 39]
- Malnutrition [40]
- Polypharmacy, ≥ 5 chronic medications [41, 42]
- Severe and persistent mental illness, cognitive impairment, or substance use disorder and, **Both:** [43, 44]
  - Patient unreliable and unable to adhere to outpatient instructions or treatment [45]
  - Caregiver unavailable or unable to manage care
- **INTERMEDIATE, ≥ One:** [46]
  - HFNC and respiratory monitoring every 3-4h [47]
  - NIPPV [48]
  - Oxygen ≥ 40%(0.40) [49]
- **CRITICAL, ≥ One:** [50]
  - Heart failure and, **Both:**
    - Finding, ≥ **One:**
      - Arterial Po_2 < 56 mmHg(7.4 kPa)
      - O_2 sat < 89%(0.89) and < baseline [9]
    - Diuretic every 1-2h or continuous [51]
  - ECMO or ECLS [52]
  - Hemodynamic monitoring, invasive [53]
  - HFNC and respiratory monitoring every 1-2h [47]
  - IABP [54]
  - Mechanical ventilation
  - NIPPV and, ≥ **One:** [48]
    - Arterial Po_2 ≤ 39 mmHg(5.2 kPa)
    - Arterial or venous Pco_2 ≥ 60 mmHg(8.0 kPa) and, **One:**
      - COPD and arterial or venous pH ≤ 7.24
      - No hx of COPD
  - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
  - VAD, percutaneous [55, 56]

Episode Day 1, One: [1, 2, 4, 5, 6]
(Symptom or finding within 24h)

- **OBSERVATION, One:** [7]
  - Atrial arrhythmia (use **Arrhythmia: Atrial** subset)
  - Heart failure, **Both:**
    - Finding, ≥ **One:**
      - Dyspnea, not returned to baseline after at least 1 dose of diuretic and ≥ 2h treatment and, ≥ **One:** [8, 9, 10]
        - O_2 sat 89-91%(0.89-0.91) and < baseline [9, 11]
        - Edema of lower extremities
        - Hepatomegaly
        - Jugular venous distention
        - Rales (crackles) on physical examination
        - Pleural effusion or pulmonary edema or cardiomegaly confirmed by CXR
        - Weight gain of ≥ 3 lbs (1.4 kg) over last 2d or ≥ 5 lbs (2.3 kg) in 7d [12]
      - Failed outpatient management, **Both:** [13]
        - Dyspnea or weight gain of ≥ 3 lbs (1.4 kg) over last 2d or ≥ 5 lbs (2.3 kg) in 7d [12]
        - ≥ 2 visits within last 7d
    - Intervention, ≥ **One:** [57]
      - Diuretic, ≥ **One:** [51, 58]
        - Diuretic ≥ 2 doses
        - Conversion from IV to PO
      - Medication adjustment (includes PO) [59]

- Dialysis
- **ACUTE, One:** [14]
  - Heart failure, new onset and, **All:** [15]
    - Symptom, ≥ **One:** [16]
      - Dyspnea
      - Orthopnea
      - Paroxysmal nocturnal dyspnea
    - Finding, ≥ **One:** [17]
      - Rales (crackles) on physical examination
      - Gallop (S3, S4) on physical examination
      - Pleural effusion or pulmonary edema or cardiomegaly confirmed by CXR
      - Edema of lower extremities
      - Hepatomegaly
      - Jugular venous distention
      - BNP or NT-proBNP > ULN [18]
    - Intervention, ≥ **One:** [58]
      - Diuretic ≥ 1 dose [51]
      - Dialysis (includes ultrafiltration)
  - Heart failure, acute on chronic and, **All:**
    - Continued symptoms after at least 1 dose of diuretic and ≥ 2h treatment and, ≥ **One:** [10, 16]
      - Dyspnea, not returned to baseline [9]
      - O_2 sat < 89%(0.89) and < baseline [9, 11]
    - Finding, ≥ **One:**
      - Inadequate diuresis, ≥ **One:** [19]
        - Urine sodium < 50 mEq/L(50 mmol/L) at 2h after diuretic [20]
        - Urine output < 150ml/h at 2h after diuretic [20]
      - Clinical risk factor, ≥ **One:** [21]
        - Troponin > ULN [22]
        - Creatinine ≥ 1.5x baseline or estimated baseline [23, 24]
        - Chronic kidney disease (excludes chronic dialysis) and creatinine ≥ 2.75 mg/dL [25]
        - BUN ≥ 43 mg/dL [26]
        - Sodium < 130 mEq/L(130 mmol/L) [27]
        - Heart rate 100-120/min, sustained [28, 29]
        - SBP < 120 mmHg and < baseline [30]
        - Frailty, **Both:** [31, 32]
          - Age ≥ 65 yrs
          - Finding, ≥ **One:**
            - Frailty, active diagnosis [33]
            - Frailty characteristics, ≥ **Two:** [34]
              - Chronic condition(s), ≥ **One:**
                - Dementia or cognitive impairment [35]
                - Multimorbidity, ≥ 2 chronic illnesses [36, 37]
              - Functional impairment [38, 39]
              - Malnutrition [40]
              - Polypharmacy, ≥ 5 chronic medications [41, 42]
        - Severe and persistent mental illness, cognitive impairment, or substance use disorder and, **Both:** [43, 44]
          - Patient unreliable and unable to adhere to outpatient instructions or treatment [45]
          - Caregiver unavailable or unable to manage care
    - Intervention, ≥ **One:**
      - Diuretic, ≥ **One:** [51, 58]
        - Diuretic ≥ 2 doses
        - Escalation or change of diuretic [60]

        - Dialysis (includes ultrafiltration)
  - **INTERMEDIATE, ≥ One:** [46]
    - Atrial arrhythmia (use **Arrhythmia: Atrial** subset)
    - Heart block (use **Arrhythmia: Blocks** subset)
    - Ventricular arrhythmia (use **Arrhythmia: Ventricular or abnormal ECG finding** subset)
    - HFNC and respiratory monitoring every 3-4h [47]
    - NIPPV [48]
    - Oxygen ≥ 40%(0.40) [49]
  - **CRITICAL, ≥ One:** [50]
    - Atrial arrhythmia (use **Arrhythmia: Atrial** subset)
    - Heart block (use **Arrhythmia: Blocks** subset)
    - Hypertension (use **Hypertension** subset)
    - Ventricular arrhythmia (use **Arrhythmia: Ventricular or abnormal ECG finding** subset)
    - Heart failure and, **Both:**
      - Finding, ≥ **One:**
        - Arterial Po_2 < 56 mmHg(7.4 kPa)
        - O_2 sat < 89%(0.89) and < baseline [9]
      - Diuretic every 1-2h or continuous [51]
    - ECMO or ECLS [52]
    - Hemodynamic monitoring, invasive [53]
    - HFNC and respiratory monitoring every 1-2h [47]
    - IABP [54]
    - Mechanical ventilation
    - NIPPV and, ≥ **One:** [48]
      - Arterial Po_2 ≤ 39 mmHg(5.2 kPa)
      - Arterial or venous Pco_2 ≥ 60 mmHg(8.0 kPa) and, **One:**
        - COPD and arterial or venous pH ≤ 7.24
        - No hx of COPD
    - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
    - VAD, percutaneous [55, 56]
- **Episode Day 2, One:** [4, 5, 61]
  - **OBSERVATION, One:**
    - **Responder**, discharge expected today if clinically stable last 12h, **All:** [62]
      - Symptoms of heart failure resolving or at baseline [9]
      - Blood pressure within acceptable limits [63]
      - Heart rate within acceptable limits [63]
      - Laboratory values normal or within acceptable limits [63]
      - O_2 sat within acceptable limits [63]
    - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
      - Persistent heart failure, **Both:**
        - Finding, ≥ **One:**
          - O_2 sat ≤ 91%(0.91) and < baseline [57, 64]
          - Signs and symptoms, ≥ **One:** [65]
            - Dyspnea or orthopnea, not returned to baseline [9]
            - Edema in lower extremities, not returned to baseline [9]
            - Jugular venous distention
            - Persistent rales
        - Intervention, **One:**
          - Diuretic, ≥ **One:** [51]
            - ≥ 2x/24h (includes PO)
            - Medication adjustment (includes PO) [66]
          - Dialysis
      - Acute deconditioning and physical therapy evaluation ≤ 24h [67]

- Abnormal vital signs ≤ 2d, **Both:**
  - Finding, ≥ **One:**
    - Heart rate < 60/min or > 100/min
    - Systolic blood pressure > 140 mmHg [68]
    - Diastolic blood pressure > 100 mmHg [68]
  - Medication adjustment (includes PO) [66]
- Hypokalemia ≤ 24h, **All:** [69]
  - Potassium < 3.0 mEq/L(3.0 mmol/L)
  - No ECG changes [70, 71]
  - Potassium repletion (includes PO)

- **ACUTE, One:**
  - **Early responder**, discharge expected today if clinically stable last 12h, **All:** [62]
    - Symptoms of heart failure resolving or at baseline [9]
    - Blood pressure within acceptable limits [63]
    - Heart rate within acceptable limits [63]
    - Laboratory values normal or within acceptable limits [63]
    - O_2 sat within acceptable limits or home oxygen therapy arranged [63]
    - No complication or active comorbidity relevant to this episode of care [72]
  - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
    - Persistent heart failure, **Both:**
      - Finding, ≥ **One:**
        - O_2 sat ≤ 91%(0.91) and < baseline [64]
        - Signs and symptoms, ≥ **One:** [73]
          - Dyspnea or orthopnea, not returned to baseline [9]
          - Edema, pre-sacral, scrotal, peroneal or lower extremities, not returned to baseline [9, 74]
          - Jugular venous distention
          - Persistent rales
      - Intervention, ≥ **One:**
        - Diuretic, ≥ **One:** [51]
          - ≥ 2x/24h (includes PO)
          - Medication adjustment (includes PO) [66]
        - Dialysis (includes ultrafiltration)
        - New onset heart failure and medication adjustment, ≤ 2d [66]
    - Cardiac catheterization scheduled or performed within 24h [75]
    - Inotrope, continuous infusion, requiring infrequent titration ≤ 2d [76, 77]
    - Acute deconditioning and physical therapy evaluation ≤ 24h [67]
    - Abnormal vital signs ≤ 2d, **Both:**
      - Finding, ≥ **One:**
        - Heart rate < 60/min or > 100/min
        - Systolic blood pressure > 140 mmHg [68]
        - Diastolic blood pressure > 100 mmHg [68]
      - Medication adjustment (includes PO) [66]
    - Blood glucose ≤ 24h, **All:** [78]
      - Finding, ≥ **One:**
        - > 250 mg/dL(13.9 mmol/L)
        - < 70 mg/dL(3.9 mmol/L)
      - Blood glucose monitoring
      - Insulin adjustment or dose held [78, 79]
    - COPD and, **Both:**
      - Persistent dyspnea, ≥ **One:** [80]
        - Arterial Po_2 < baseline [9, 81]
        - Arterial or venous Pco_2 > baseline [9, 81]
        - O_2 sat ≤ 91%(0.91) and < baseline [9]

                - Increased work of breathing, ≥ **One:**
                    - Difficulty taking PO
                    - Prefers sitting [82]
                    - Talks in phrases
                    - Use of accessory muscles
            - Intervention, **Both:**
                - Bronchodilator ≥ 4x/24h [83, 84]
                - Corticosteroid (includes PO or inhaled) [85]
        - Hypokalemia, **All:** [69]
            - Potassium < 3.0 mEq/L(3.0 mmol/L)
            - No ECG changes [70, 71]
            - Potassium repletion (includes PO)
        - Hyponatremia or SIADH, **Both:** [86]
            - Sodium 120-129 mEq/L(120-129 mmol/L)
            - Intervention, ≥ **One:**
                - Diuretic and oral sodium supplement
                - Fluid restriction
                - Medication, ≥ **One:**
                    - Demeclocycline (includes PO)
                    - Diuretic ≥ 2x/24h
                - Sodium chloride 0.9%(0.009) [87]
        - Pneumonia confirmed by imaging and, **Both:** [88]
            - Finding, ≥ **One:**
                - Temperature ≥ 100.4°F(38.0°C) [89]
                - Mental status change (excludes coma, stupor, or obtundation) or GCS 9-14 [90]
                - O_2 sat ≤ 91%(0.91) and < baseline [9]
                - WBC ≥ 12,000/cu.mm(12x10^9/L)
                - WBC ≤ 4,000/cu.mm(4 x10^9/L)
                - Bands > 10%(0.10)
            - Intervention, ≥ **One:** [91]
                - Requiring supplemental oxygen [49]
                - Anti-infective (includes PO)
                - Respiratory interventions at least 6x/24h [92]
        - Post Critical care ≤ 24h
- **INTERMEDIATE,** ≥ **One:**
    - HFNC and respiratory monitoring every 3-4h [47]
    - Hypokalemia, **All:** [69]
        - Potassium < 3.0 mEq/L(3.0 mmol/L)
        - PVCs > 6/min
        - Potassium repletion
    - NIPPV [48, 92]
    - Oxygen ≥ 40%(0.40), ≤ 2d [49]
- **CRITICAL,** ≥ **One:** [93]
    - ACS known or suspected (use **Acute Coronary Syndrome** subset)
    - Heart failure, **Both:**
        - Finding, ≥ **One:**
            - Arterial Po_2 < 56 mmHg(7.4 kPa)
            - O_2 sat < 89%(0.89) and < baseline [9]
        - Diuretic every 1-2h or continuous [51]
    - ECMO or ECLS [52]
    - Hemodynamic monitoring, invasive [53]
    - HFNC and respiratory monitoring every 1-2h [47]
    - Hypokalemia, **All:** [69, 94]

- Potassium < 3.0 mEq/L(3.0 mmol/L)
- ⊖ ECG findings, ≥ **One:** (94)
  - Multifocal PVCs
  - R on T phenomenon (95)
  - T and U wave fusion (96)
  - U wave (97)
- Potassium repletion (69)

⊖ Hyponatremia, **Both:**
- Sodium < 120 mEq/L(120 mmol/L) and symptomatic (98, 99)
- ⊖ Intervention, ≥ **One:**
  - Conivaptan or tolvaptan (includes PO) (100)
  - Diuretic ≥ 2x/24h
  - Fluid restriction
  - Sodium chloride 3%(0.03) (87)

- IABP (54)
- Mechanical ventilation
- ⊖ NIPPV and, ≥ **One:** (48)
  - FiO_2 > 50%(0.50)
  - Respiratory intervention every 1-2h (92)
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- VAD, percutaneous (55, 56)

⊖ **Episode Day 3, One:** (5, 101, 102)

⊖ **OBSERVATION, One:**

⊖ **Responder,** discharge expected today if clinically stable last 12h, **All:** (62)
- Symptoms of heart failure resolving or at baseline (9)
- Blood pressure within acceptable limits (63)
- Heart rate within acceptable limits (63)
- Laboratory values normal or within acceptable limits (63)
- O_2 sat within acceptable limits (63)

⊖ **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
- For heart failure use Episode Day 3 at a higher level of care
- For a condition other than heart failure use appropriate subset (Episode Day 1) based on clinical findings
- Refer for secondary review

⊖ **ACUTE, One:**

⊖ **Responder,** discharge expected today if clinically stable last 12h, **All:** (62)
- Afebrile
- Symptoms of heart failure resolving or at baseline (9)
- Blood pressure within acceptable limits (63, 68)
- Heart rate within acceptable limits (63)
- O_2 sat within acceptable limits or home oxygen therapy arranged (63)
- ⊖ Complication or comorbidity, **One:**
  - No complication or active comorbidity relevant to this episode of care (72)
  - ⊖ Complication or comorbidity and clinically stable for discharge, ≥ **One:**
    - Blood glucose within acceptable limits (63)
    - COPD exacerbation resolving
    - ⊖ Deconditioned, **One:** (103, 104)
      - Discharge to home and patient or caregiver able to safely manage impairment or home care services arranged
      - Discharge to inpatient facility for rehabilitative services
    - Electrolytes within acceptable limits
    - WBC within acceptable limits (63, 105)

⊖ **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**

- Persistent heart failure, **Both:**
  - Finding, ≥ **One:**
    - O_2 sat ≤ 91%(0.91) and < baseline [57, 64]
    - Signs and symptoms, ≥ **One:** [73]
      - Dyspnea or orthopnea, not returned to baseline [9]
      - Edema, pre-sacral, scrotal, peroneal or lower extremities, not returned to baseline [9, 74]
      - Jugular venous distention
      - Persistent rales
  - Intervention, ≥ **One:**
    - Diuretic, ≥ **One:** [51]
      - ≥ 2x/24h (includes PO)
      - Requires daily medication adjustment (includes PO) and BUN or creatinine increased from baseline [9, 66]
    - Dialysis (includes ultrafiltration)
    - New onset heart failure and medication adjustment, ≤ 2d [66]
- Cardiac catheterization scheduled or performed within 24h [75]
- Inotrope, continuous infusion, requiring infrequent titration ≤ 2d [76, 77]
- Acute deconditioning and physical therapy evaluation ≤ 24h [67]
- Abnormal vital signs ≤ 2d, **Both:**
  - Finding, ≥ **One:**
    - Heart rate < 60/min or > 100/min
    - Systolic blood pressure > 140 mmHg [68]
    - Diastolic blood pressure > 100 mmHg [68]
  - Medication adjustment (includes PO) [66]
- Blood glucose ≤ 24h, **All:** [78]
  - Finding, ≥ **One:**
    - > 250 mg/dL(13.9 mmol/L)
    - < 70 mg/dL(3.9 mmol/L)
  - Blood glucose monitoring
  - Insulin adjustment or dose held [78, 79]
- COPD and, **Both:**
  - Persistent dyspnea, ≥ **One:** [80]
    - Arterial Po_2 < baseline [9, 81]
    - Arterial or venous Pco_2 > baseline [9, 81]
    - O_2 sat ≤ 91%(0.91) and < baseline [9]
    - Increased work of breathing, ≥ **One:**
      - Difficulty taking PO
      - Prefers sitting [82]
      - Talks in phrases
      - Use of accessory muscles
  - Intervention, **Both:**
    - Bronchodilator ≥ 4x/24h [83, 84]
    - Corticosteroid (includes PO or inhaled) [85]
- Hypokalemia, **All:** [69]
  - Potassium < 3.0 mEq/L(3.0 mmol/L)
  - No ECG changes [70, 71]
  - Potassium repletion (includes PO)
- Hyponatremia or SIADH, **Both:** [86]
  - Sodium 120-129 mEq/L(120-129 mmol/L)
  - Intervention, ≥ **One:**
    - Diuretic and oral sodium supplement
    - Fluid restriction
    - Medication, ≥ **One:**

- Demeclocycline (includes PO)
- Diuretic ≥ 2x/24h
- Sodium chloride 0.9%(0.009) (87)
- Pneumonia confirmed by imaging and, **Both:** (88)
  - Finding, ≥ **One:**
    - Temperature ≥ 100.4°F(38.0°C) (89)
    - Mental status change (excludes coma, stupor, or obtundation) or GCS 9-14 (90)
    - O_2 sat ≤ 91%(0.91) and < baseline (9)
    - WBC ≥ 12,000/cu.mm(12x10^9/L)
    - WBC ≤ 4,000/cu.mm(4 x10^9/L)
    - Bands > 10%(0.10)
  - Intervention, ≥ **One:** (91)
    - Requiring supplemental oxygen (49)
    - Anti-infective (includes PO)
    - Respiratory interventions at least 6x/24h (92)
- Post Critical care ≤ 24h

- **INTERMEDIATE,** ≥ **One:**
  - HFNC and respiratory monitoring every 3-4h (47)
  - Hypokalemia, **All:** (69)
    - Potassium < 3.0 mEq/L(3.0 mmol/L)
    - PVCs > 6/min
    - Potassium repletion
  - NIPPV (48, 92)
  - Oxygen ≥ 40%(0.40), ≤ 2d (49)
- **CRITICAL,** ≥ **One:** (93)
  - Heart failure, **Both:**
    - Finding, ≥ **One:**
      - Arterial Po_2 < 56 mmHg(7.4 kPa)
      - O_2 sat < 89%(0.89) and < baseline (9)
    - Diuretic every 1-2h or continuous (51)
  - ECMO or ECLS (52)
  - Hemodynamic monitoring, invasive (53)
  - HFNC and respiratory monitoring every 1-2h (47)
  - Hypokalemia, **All:** (69, 94)
    - Potassium < 3.0 mEq/L(3.0 mmol/L)
    - ECG findings, ≥ **One:** (94)
      - Multifocal PVCs
      - R on T phenomenon (95)
      - T and U wave fusion (96)
      - U wave (97)
    - Potassium repletion (69)
  - Hyponatremia, **Both:**
    - Sodium < 120 mEq/L(120 mmol/L) and symptomatic (98, 99)
    - Intervention, ≥ **One:**
      - Conivaptan or tolvaptan (includes PO) (100)
      - Diuretic ≥ 2x/24h
      - Fluid restriction
      - Sodium chloride 3%(0.03) (87)
  - IABP (54)
  - Mechanical ventilation
  - NIPPV and, ≥ **One:** (48)
    - FiO_2 > 50%(0.50)
    - Respiratory intervention every 1-2h (92)

    - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
    - VAD, percutaneous [55, 56]
- **Episode Day 4, One:** [5, 102]
  - **ACUTE, One:**
    - **Responder,** discharge expected today if clinically stable last 12h, **All:** [62]
      - Afebrile
      - Symptoms of heart failure resolving or at baseline [9]
      - Blood pressure within acceptable limits [63, 68]
      - Heart rate within acceptable limits [63]
      - O_2 sat within acceptable limits or home oxygen therapy arranged [63]
      - Complication or comorbidity, **One:**
        - No complication or active comorbidity relevant to this episode of care [72]
        - Complication or comorbidity and clinically stable for discharge, ≥ **One:**
          - Blood glucose within acceptable limits [63]
          - COPD exacerbation resolving
          - Deconditioned, **One:** [103, 104]
            - Discharge to home and patient or caregiver able to safely manage impairment or home care services arranged
            - Discharge to inpatient facility for rehabilitative services
          - Electrolytes within acceptable limits
          - WBC within acceptable limits [63, 105]
    - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
      - Persistent heart failure, **Both:**
        - Finding, ≥ **One:**
          - O_2 sat ≤ 91%(0.91) and < baseline [57, 64]
          - Signs and symptoms, ≥ **One:** [73]
            - Dyspnea or orthopnea, not returned to baseline [9]
            - Edema, pre-sacral, scrotal, peroneal or lower extremities, not returned to baseline [9, 74]
            - Jugular venous distention
            - Persistent rales
        - Intervention, ≥ **One:**
          - Diuretic, ≥ **One:** [51]
            - ≥ 2x/24h (includes PO)
            - Requires daily medication adjustment (includes PO) and BUN or creatinine increased from baseline [9, 66]
          - Dialysis (includes ultrafiltration)
          - New onset heart failure and medication adjustment, ≤ 2d [66]
      - Cardiac catheterization scheduled or performed within 24h [75]
      - Inotrope, continuous infusion, requiring infrequent titration ≤ 2d [76, 77]
      - Acute deconditioning and physical therapy evaluation ≤ 24h [67]
      - Abnormal vital signs ≤ 2d, **Both:**
        - Finding, ≥ **One:**
          - Heart rate < 60/min or > 100/min
          - Systolic blood pressure > 140 mmHg [68]
          - Diastolic blood pressure > 100 mmHg [68]
        - Medication adjustment (includes PO) [66]
      - Blood glucose ≤ 24h, **All:** [78]
        - Finding, ≥ **One:**
          - > 250 mg/dL(13.9 mmol/L)
          - < 70 mg/dL(3.9 mmol/L)
        - Blood glucose monitoring
        - Insulin adjustment or dose held [78, 79]
      - COPD and, **Both:**

- Persistent dyspnea, ≥ **One:** [80]
  - Arterial Po_2 < baseline [9, 81]
  - Arterial or venous Pco_2 > baseline [9, 81]
  - O_2 sat ≤ 91%(0.91) and < baseline [9]
  - Increased work of breathing, ≥ **One:**
    - Difficulty taking PO
    - Prefers sitting [82]
    - Talks in phrases
    - Use of accessory muscles
- Intervention, **Both:**
  - Bronchodilator ≥ 4x/24h [83, 84]
  - Corticosteroid (includes PO or inhaled) [85]

- Hypokalemia, **All:** [69]
  - Potassium < 3.0 mEq/L(3.0 mmol/L)
  - No ECG changes [70, 71]
  - Potassium repletion (includes PO)
- Hyponatremia or SIADH, **Both:** [86]
  - Sodium 120-129 mEq/L(120-129 mmol/L)
  - Intervention, ≥ **One:**
    - Diuretic and oral sodium supplement
    - Fluid restriction
    - Medication, ≥ **One:**
      - Demeclocycline (includes PO)
      - Diuretic ≥ 2x/24h
    - Sodium chloride 0.9%(0.009) [87]
- Pneumonia confirmed by imaging and, **Both:** [88]
  - Finding, ≥ **One:**
    - Temperature ≥ 100.4°F(38.0°C) [89]
    - Mental status change (excludes coma, stupor, or obtundation) or GCS 9-14 [90]
    - O_2 sat ≤ 91%(0.91) and < baseline [9]
    - WBC ≥ 12,000/cu.mm($12x10^9$/L)
    - WBC ≤ 4,000/cu.mm($4 x10^9$/L)
    - Bands > 10%(0.10)
  - Intervention, ≥ **One:** [91]
    - Requiring supplemental oxygen [49]
    - Anti-infective (includes PO)
    - Respiratory interventions at least 6x/24h [92]
- Post Critical care ≤ 24h

- **INTERMEDIATE, ≥ One:**
  - HFNC and respiratory monitoring every 3-4h [47]
  - Hypokalemia, **All:** [69]
    - Potassium < 3.0 mEq/L(3.0 mmol/L)
    - PVCs > 6/min
    - Potassium repletion
  - NIPPV [48, 92]
  - Oxygen ≥ 40%(0.40), ≤ 2d [49]
- **CRITICAL, ≥ One:** [93]
  - Heart failure, **Both:**
    - Finding, ≥ **One:**
      - Arterial Po_2 < 56 mmHg(7.4 kPa)
      - O_2 sat < 89%(0.89) and < baseline [9]
    - Diuretic every 1-2h or continuous [51]
  - ECMO or ECLS [52]

- Hemodynamic monitoring, invasive [53]
- HFNC and respiratory monitoring every 1-2h [47]
- Hypokalemia, **All:** [69, 94]
  - Potassium < 3.0 mEq/L(3.0 mmol/L)
  - ECG findings, ≥ **One:** [94]
    - Multifocal PVCs
    - R on T phenomenon [95]
    - T and U wave fusion [96]
    - U wave [97]
  - Potassium repletion [69]
- Hyponatremia, **Both:**
  - Sodium < 120 mEq/L(120 mmol/L) and symptomatic [98, 99]
  - Intervention, ≥ **One:**
    - Conivaptan or tolvaptan (includes PO) [100]
    - Diuretic ≥ 2x/24h
    - Fluid restriction
    - Sodium chloride 3%(0.03) [87]
- IABP [54]
- Mechanical ventilation
- NIPPV and, ≥ **One:** [48]
  - FiO_2 > 50%(0.50)
  - Respiratory intervention every 1-2h [92]
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- VAD, percutaneous [55, 56]

Episode Day 5, One:

- **ACUTE, One:**
  - **Responder,** discharge expected today if clinically stable last 12h, **All:** [62]
    - Afebrile
    - Symptoms of heart failure resolving or at baseline [9]
    - Blood pressure within acceptable limits [63, 68]
    - Heart rate within acceptable limits [63]
    - O_2 sat within acceptable limits or home oxygen therapy arranged [63]
    - Complication or comorbidity, **One:**
      - No complication or active comorbidity relevant to this episode of care [72]
      - Complication or comorbidity and clinically stable for discharge, ≥ **One:**
        - Blood glucose within acceptable limits [63]
        - COPD exacerbation resolving
        - Deconditioned, **One:** [103, 104]
          - Discharge to home and patient or caregiver able to safely manage impairment or home care services arranged
          - Discharge to inpatient facility for rehabilitative services
        - Electrolytes within acceptable limits
        - WBC within acceptable limits [63, 105]
  - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
    - **Use Extended Stay subset**
    - Refer for secondary review
- **INTERMEDIATE, One:**
  - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**
    - **Use Extended Stay subset**
    - Refer for secondary review
- **CRITICAL, One:**

- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
  - **Use Extended Stay subset**
  - Refer for secondary review

Discharge, Level of Care, One: [106]

- **HOME, All:**
  - Level of care appropriateness, **Both:**
    - Home environment safe and accessible [107]
    - Patient or caregiver demonstrates ability to manage care
  - Heart failure discharge planning, **Both:** [108]
    - Provide comprehensive written discharge and teaching instructions [109]
    - Education on the importance of, **All:** [110]
      - Strict medication adherence
      - Monitoring sodium intake
      - Monitoring daily weights and reporting a gain of > 2 lbs/day to their medical practitioner
      - Participating in an exercise program
      - Avoiding tobacco, excessive alcohol, illicit drug, and nonsteroidal anti-inflammatory drug (NSAID) use
      - Understanding when and where to seek help
  - Medication reconciliation [111]
  - Follow-up care planned [112, 113]
  - Identify and address transportation needs
- **HOME CARE, All:**
  - Level of care appropriateness, **All:**
    - Home environment safe and accessible [107]
    - Patient or caregiver willing and able to learn care
    - Outpatient management contraindicated or unavailable [114, 115]
    - Skilled services, ≥ **One:** [116]
      - Skilled nursing [117]
      - Skilled therapy, PT, OT, or ST
      - Behavioral health [118]
  - Heart failure discharge planning, **Both:** [108]
    - Provide comprehensive written discharge and teaching instructions [109]
    - Education on the importance of, **All:** [110]
      - Strict medication adherence
      - Monitoring sodium intake
      - Monitoring daily weights and reporting a gain of > 2 lbs/day to their medical practitioner
      - Participating in an exercise program
      - Avoiding tobacco, excessive alcohol, illicit drug, and nonsteroidal anti-inflammatory drug (NSAID) use
      - Understands when and where to seek help
  - Medication reconciliation [111]
  - Follow-up care planned [113]
  - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
  - Level of care appropriateness, **One:**
    - Support systems available [119]
  - Skilled treatment, ≥ **One:**
    - Clinical assessment [120]
    - Individual, group, or family counseling
    - Psychiatric, substance, or medication evaluation [121]
- **SKILLED MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk

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- Treatment precluded in a lower level
- Services, ≥ **One:**
  - Medical and skilled nursing interventions or assessment 1-2x/24h [117]
  - Therapy, **All:**
    - Goal directed therapy with rehabilitation potential [122, 123]
    - Functional impairments requiring at least supervision [124]
    - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
- Complete prior to facility transfer, **All:**
  - Medication reconciliation [111]
  - Obtain and complete forms for facility
  - Obtain discharge summary and transmit to facility and medical practitioner
  - Arrange transportation

- **SUBACUTE MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 4h/24h [117]
      - Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [122, 123]
        - ≥ 2 functional impairments requiring at least minimum assistance [124]
        - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [111]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **SUBACUTE REHAB, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Treatment precluded in a lower level
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential [122, 123]
      - ≥ 2 functional impairments requiring at least minimum assistance [124]
      - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [111]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **ACUTE REHAB, Both:**
  - Level of care appropriateness, **Both:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential [122, 123]
      - ≥ 2 functional impairments requiring at least minimum assistance [124]
      - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines [125]
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [111]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**

- Level of care appropriateness, **Both:**
  - Treatment precluded in a lower level
  - In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:**
    - Primary condition or illness and treatment of active comorbid condition(s) [126]
    - Ventilator dependent ≥ 6h/d and weaning planned [127, 128]
    - Acute and comorbid conditions requiring prolonged hospitalization [129]
- Complete prior to facility transfer, **All:**
  - Medication reconciliation [111]
  - Obtain and complete forms for facility
  - Obtain discharge summary and transmit to facility and medical practitioner
  - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
291 HEART FAILURE AND SHOCK WITH MCC	**n/a**	**3.9**	**CMS GMLOS**
292 HEART FAILURE AND SHOCK WITH CC	**n/a**	**3**	**CMS GMLOS**
293 HEART FAILURE AND SHOCK WITHOUT CC/MCC	**n/a**	**2.1**	**CMS GMLOS**
HEART FAILURE	**30%**	**4.8**	**InterQual**

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:

1:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.

- **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Transition Plan: InterQual's® Transition Plan identifies patients at high risk for readmission who may benefit from a comprehensive discharge plan. Heart failure (HF) ranks in the top 20 diagnoses with the highest 7-day and 30-day readmission rates. The 30-day all-cause readmission rate is as high as 23%(0.23) (Fingar et al., In: Healthcare Cost and Utilization Project (HCUP) Statistical Briefs. 2017). HF is a common condition in skilled nursing facilities (SNFs), with cardiovascular diagnoses as the largest diagnostic category for this setting. HF patients account for up to 27% (0.27) to 43%(0.43) of SNF 30-day rehospitalization rates (Jurgens et al., J Card Fail 2015, 21: 263-99). HF patients, in general, identified to be at highest risk for readmission include those with:

- Moderate to severe HF and age 65 or older (Iyngkaran et al., Clinical Medicine Insights: Cardiology 2018, 12: 1179546818809358)
- Age less than 65 at the time of initial HF admission (Iyngkaran et al., Clinical Medicine Insights: Cardiology 2018, 12: 1179546818809358)
- Age 65 or older and post major noncardiac surgery (Turrentine et al., J Am Coll Surg 2016, 222: 1220-9)
- Patients with preexisting atrial fibrillation (Tripathi et al., J Am Heart Assoc 2019, 8: e013026)

- Existence of a comorbid condition including diabetes, renal failure, chronic pulmonary disease, anemia, depression, and fluid and electrolyte disorder (Chamberlain et al., Int J Gen Med 2018, 11: 127-41; Arora et al., Am J Cardiol 2017, 120: 616-24)
- High pre-discharge B-type natriuretic peptide (BNP) level and less than 50%(0.50) decrease from admission level
- Poor comprehension of discharge instruction related to limited educational background or primary language other than English (Iyngkaran et al., Clinical Medicine Insights: Cardiology 2018, 12: 1179546818809358)
- Poor medication adherence (Ruppar et al., J Am Heart Assoc 2016, 5:)
- Medicaid coverage (Iyngkaran et al., Clinical Medicine Insights: Cardiology 2018, 12: 1179546818809358)
- Longer initial hospital stays (Arora et al., Am J Cardiol 2017, 120: 616-24; Albert et al., Circ Heart Fail 2015, 8: 384-409)
- Multiple emergency department visits within 6 months of hospitalization (Albert et al., Circ Heart Fail 2015, 8: 384-409)

A BNP level of greater than 350 pg/mL or less than a 50%(0.50) reduction in N-terminal prohormone BNP (NT-proBNP) during the hospital stay is also associated with an increased risk for rehospitalization or death, as is the development of hypotension during hospitalization (Patel et al., Circ Heart Fail 2014, 7: 918-25). Patients with a discharge BNP ≥ 1000 pg/mL had an unadjusted 30-day HF-specific readmission rate over 3 times as high as patients whose discharge BNP was ≤ 200 pg/mL (Flint et al., J Am Heart Assoc 2014, 3: e000806).

3:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

4:

Guideline Standard Considerations

- Beta blocker if not contraindicated
- Angiotensin converting enzyme inhibitors (ACEi), angiotensin receptor blockers (ARB), or angiotensin receptor neprilysin inhibitors (ARNi) if not contraindicated
- Continuous cardiac monitoring
- Deep venous thrombosis prophylaxis

Expected progress after 24h

- Dyspnea improving
- O_2 saturation improving
- Activity increasing
- Vital signs stabilizing

Consider:

- Decreasing supplemental O_2
- Changing medications to PO
- Nutrition consult for sodium restriction
- Discharge planning

5:

Care Facilitation

Prolonged oxygen therapy expected

- Home oxygen therapy should be considered in a clinically stable patient who continues to require supplemental oxygen when there has been no change in oxygen requirements within last 2 days **(Home)**

Prolonged inotrope therapy expected

- Inotrope therapy should be considered in a clinically stable patient who continues to require support at a stable dose **(Home Care, SAC)**

Potential risk of readmission

- Knowledge deficit or non-adherence with condition management identified **(Home Care, Readmission reduction team)**
- Co-morbid chronic obstructive pulmonary disease, diabetes, chronic kidney disease, atrial fibrillation **(Home Care, Readmission reduction team)**
- Heart Failure admission within the last 6 months **(Home Care, Readmission reduction team)**

Functional or activity of daily living (ADL) impairment - consider an alternate level of care for therapy needs

- Physical therapy provided in the home setting should be considered if the home is safe and accessible **(Home Care, Outpatient Cardiac Rehabilitation)**
- If home environment is unsafe, consider skilled nursing facility **(SNF)**

Prolonged ventilation and unable to wean

- Prolonged ventilator weaning expected - consider transfer to a long term acute care (LTAC) for failed mechanical ventilation weaning longer than 1 week **(SAC, LTAC)**

End of life planning

- Consider Palliative care for patients for end stage disease
- Consider hospice care for terminally ill patients with a life expectancy of less than 6 months

Social Determinants of Health (SDOH)

Social concerns may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. The following social determinants of health should be evaluated on admission to determine which factors are relevant to an individual patient and condition. This list may not be all inclusive and all relevant SDOH factors should be considered. Any SDOH factors determined to be relevant to a given patient should be planned for throughout the hospital stay.

Alcohol or substance use

- Care coordination with outpatient medical or behavioral health providers
- Information should be given about community resources and programs
- Referral to addiction social support systems

Domestic violence and sexually abused individuals

- Community resources and legal information should be given
- Create a long-term safety plan or make appropriate referral

End of life planning, arrange palliative care or hospice

- Consider palliative care for patients with end stage disease
- Consider hospice care for terminally ill patients with a life expectancy of less than 6 months

Financial resources strained or limited

- For patients with income insecurity, identify income-support resources (Pottie et al., CMAJ 2020, 192: E240-E54)
- Consider direct income assistance benefits and indirect income assistance programs (i.e., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)
- Consider referral to Area Agencies on Aging (AAA) for information on nutrition and meals programs, home homemaking services, transportation resources
- For food insecurity, provide information about local food banks and Supplemental Nutrition Assistance

Program (SNAP) incentive programs (Berman et al., Pediatr Rev 2018, 39: 235-46)

Functional or ADL impairment

- Consider an alternate level of care
- If patient is unable to make decision for self, refuses treatment, or demands to be discharged without a rational reason, a capacity evaluation may be needed to ensure patient's ability to make decisions for self
- Information should be given to patient and family about Advance Directive or Power of Attorney

Mental health co-morbidities involved

- Appointment with primary care provider or behavioral health specialist upon discharge
- Care coordination with outpatient behavioral health providers
- Community resources should be given

Risk for readmission

- Consider home care services
- Consider transfer to an alternative level of care or readmission reduction program
- Provide patient and family with information on Disease Management programs
- Consider telehealth and mobile health interventions

Risk for treatment nonadherence

- For patients with poor insight into illness or noncompliance with treatment, introduce peer support groups to help improve service user experience and quality of life (National Institute for Health and Care Excellence (NICE). Psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2014.)
- In-home intervention to prevent medical relapse, or readmission, and to enhance medication compliance
- For patients at risk of medication nonadherence, consider patient education and behavioral support initiatives such as text message reminders to take a medication and electronic medication dispensers

Environmental exposure to toxic substances and other physical hazards

- For tobacco use and/or exposure, information should be given about tobacco-use cessation and the risk of second-hand smoke
- For lead poisoning, provide community resources for primary prevention and secondary prevention strategies (i.e., blood lead level testing and follow up) (Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019)

Unstable living environment or homelessness expected upon discharge

- Health insurance application
- Homeless shelter application and information should be given to patient
- Referral to walk-in state or federal sponsored outpatient programs
- Ensure access to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., CMAJ 2020, 192: E240-E54)
- For individuals with homelessness and the greatest service need (i.e., serious mental illness, history of hospitalizations, functional impairment), consider Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community-Based Treatment (Ponka et al., PLoS One 2020, 15: e0230896)

Unsafe home environment

- If home environment is unsafe, consider a skilled nursing facility (SNF)
- For patients at risk for falls, provide information on community-based fall prevention programs through the National Council on Aging

6:

Prior to initiating a review, obtain documentation of the following information in order to apply the criteria or provide additional information for a possible secondary review:

Clinical findings following at least 2h of diuretic therapy, including:

- O_2 saturation
- Heart rate
- Blood urea nitrogen level

- Creatinine level
- Edema
- Hepatomegaly
- Jugular venous distention
- Recent weight gain
- Ejection fraction, if known
- Previous outpatient therapy

Any active comorbid conditions or risk factors, such as:

- Angina
- Cardiac troponin
- Chronic kidney disease
- COPD
- Diabetes mellitus
- Pneumonia
- Mental illness
- Hyponatremia

Significant social or safety concerns, such as:

- The absence of a willing or able caregiver, if required
- An unsafe home environment
- Access to medications and follow-up care
- Access to transportation
- Non-compliance with treatments
- Frequent ED and/or hospital admissions

7:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

8:

Physical examination findings such as new onset lower extremity edema, jugular venous distention, hepatomegaly, and sudden weight gain can be characteristics of decompensation in patients with heart failure (Hollenberg et al., J Am Coll Cardiol 2019, 74: 1966-2011; Peacock et al., Ann Emerg Med 2006, 47: 22-33).

9:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

10:

Treatment of congestive heart failure (HF) is focused on hemodynamic stabilization, support of oxygenation or ventilation, and decongestion. Medications, including intravenous (IV) loop diuretics to promptly treat fluid overload, and oxygen administration are the cornerstone of initial management. The onset of action of IV loop diuretic occurs within minutes and urine output usually increases measurably within 2 hours (Heidenreich et al., Circulation 2022, 145: e895-e1032; McDonagh et al., Eur Heart J 2021, 42: 3599-726; Felker et al., Journal of the American College of Cardiology 2020, 75: 1178-95).

11:

Instruction: Patients with low oxygen saturations requiring high flow oxygen, noninvasive positive pressure ventilation, or mechanical ventilation should be considered for a higher level of care.

12:

Many guidelines include daily weight monitoring as part of self-monitoring instruction after hospital discharge, but no specific weight parameters are given as indicators for hospitalization, and available literature have conflicting data. The European Society of Cardiology states that diuretics may need to be increased with unexplained weight gain of more than 2 kg (4.4 lbs) in 3 days. A case-control study with 268 patients (134 case patients and 134 matched control patients) noticed that a weight gain of more than 2 lbs (0.9 kg) in 7 days was associated with increased risk of hospitalization for heart failure (HF). This association was stronger with weight gain of more than 5 lbs (2.3 kg) in 7 days (odds ratio of 4.46) and with weight gain of more than 10 lbs (4.5 kg) in 7 days (odds ratio of 7.65) (Chaudhry et al., Circulation 2007, 116: 1549-54). However, a more recent study, a secondary analysis of randomized control trial, suggested that rapid weight gain of 5 lbs (2.3 kg) in 7 days was associated with an increased risk of emergency department visit, but not necessarily with hospitalization (Howie-Esquivel et al., ESC Heart Fail 2019, 6: 131-7). InterQual® external peer reviewers agree that it is reasonable to monitor patients with weight gain of 3 lbs (1.4kg) or more over 2 days or 5 lbs (2.3kg) or more in 7 days in the Observation level of care.

13:

These criteria refer to the lack of response to treatment received through the medical practitioner's office, outpatient clinic (e.g., urgent care), teleconsultation, trial of home care services, or the emergency department (ED). These criteria can be applied when the patient presents with persistent heart failure (HF) symptoms despite recent visits to medical professionals (within last 7 days) in attempt to relieve their symptoms.

14:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

15:

The American College of Cardiology Expert Consensus suggests that patients presenting with new onset heart failure (HF) and signs of congestion should be admitted to the hospital for an evaluation of underlying etiologies (e. g., ischemia, valvular heart disease) and monitored initiation of treatment (Hollenberg et al., J Am Coll Cardiol 2019, 74: 1966-2011). Treatment regimen may include a combination of diuretics, vasodilators, guideline-directed medical therapy (GDMT), which includes angiotensin-converting enzyme inhibitors (ACEi) or angiotensin receptor blockers (ARB) or angiotensin receptor neprilysin inhibitors (ARNi), mineralocorticoid receptor antagonists (MRA), and sodium-glucose cotransporter 2 inhibitors (SGLT2i). Daily changes in dose, frequency, or medication may be required for adequate treatment (Heidenreich et al., Circulation 2022, 145: e895-e1032; Maddox et al., J Am Coll Cardiol 2021, 77: 772-810).

16:

When congestion is not completely relieved after initial emergency department treatment and symptoms persist, additional decongestion is necessary until symptoms have resolved. The National Heart, Lung, and Blood Institute-sponsored trials of acute decompensated heart failure (ADHF) show that rehospitalization and death are higher in patients that do not have complete symptom relief (Heidenreich et al., Circulation 2022, 145: e895-e1032; Hollenberg et al., J Am Coll Cardiol 2019, 74: 1966-2011).

17:

Many patients presenting with decompensated heart failure will have complaints of symptoms along with at least one of the following signs (Hollenberg et al., J Am Coll Cardiol 2019, 74: 1966-2011):

- Jugular venous distention
- Rales (crackles) on lung examination
- Pleural effusion
- Pulmonary edema
- Cardiomegaly
- Third heart sound

- Murmur
- Hepatomegaly
- Peripheral edema of lower extremities

18:

Natriuretic biomarkers, B-type natriuretic peptide (BNP) and N-terminal prohormone of B-type natriuretic peptide (NT-proBNP), are frequently used to evaluate patients with possible, or known, heart failure (HF). In symptomatic patients, high BNP and NT-proBNP levels can help in diagnosing HF. When the cause of dyspnea is unclear, low levels of BNP and NT-proBNP may be useful to rule out HF. However, there are other conditions that can be associated with elevated BNP and NT-proBNP levels such as acute coronary syndrome, pericardial disease, atrial fibrillation, renal failure, sepsis, and many other cardiac and noncardiac conditions. Obesity can lower natriuretic biomarkers. Higher levels of BNP and NT-proBNP in HF patients have been associated with higher risk of all-cause and cardiovascular death and major cardiovascular adverse outcomes (Heidenreich et al., Circulation 2022, 145: e895-e1032).

19:

In patients presenting with acute heart failure (HF) and evidence of fluid overload (by symptoms, physical examination, or radiologic evaluation), inadequate response to the initial intravenous (IV) dose of a loop diuretic indicates a need for the intensification of therapy in order to achieve decongestion and improvement of symptoms. The onset of action of IV loop diuretic occurs within minutes and urine output usually increases measurably within 2 hours. A spot urine sodium less than 50 mmol/L or urine output of less than 150ml per hour at 2 hours (or less than 100-150 ml per hour during first 6 hours) after the administration of a diuretic is considered an inadequate response to treatment (McDonagh et al., Eur Heart J 2021, 42: 3599-726; Felker et al., Journal of the American College of Cardiology 2020, 75: 1178-95; Mullens et al., Eur J Heart Fail 2019, 21: 137-55). A systematic review supported measuring spot urine sodium to determine which patients exhibit inadequate diuretic response (Tersalvi et al., Eur Heart J Acute Cardiovasc Care 2021, 10: 216-23). A post-hoc analysis of the ROSE Acute Heart Failure randomized trial found that a spot urine sodium of less than or equal to 60 mmol/L in patients with acute HF treated with diuretics predicted lower weight loss with diuresis and a prolonged hospital length of stay (Cunningham et al., Clin Cardiol 2020, 43: 43-9).

20:

Instruction: Diuretic here refers to the IV dose given in the emergency room or urgent care prior to decision to admit.

21:

Patients with an increased blood urea nitrogen (BUN), low systolic blood pressure, or elevated creatinine need close monitoring and treatment as these are known markers for poor outcomes in patients hospitalized with decompensated heart failure (Singer Fisher et al., Emerg Med Pract 2017, 19: S1-s2; Elkayam et al., Am Heart J 2007, 153: 98-104; Peacock et al., Ann Emerg Med 2006, 47: 22-33).

22:

Increased troponin levels with decompensated heart failure (HF) are linked to poorer outcomes during a hospitalization and within 90 days post discharge. Because of the increased mortality risk, these patients may require admission for treatment, close monitoring, and evaluation for underlying ischemia (Hollenberg et al., J Am Coll Cardiol 2019, 74: 1966-2011; Singer Fisher et al., Emerg Med Pract 2017, 19: S1-s2)

23:

When the baseline serum creatinine level is unknown, the Modification of Diet in Renal Disease (MDRD) equation can be used to estimate an expected creatinine value (assuming a normal baseline value for glomerular filtration rate (GFR) of 75ml/min per 1.73 meters squared). The following values demonstrate one method to estimate baseline serum creatinine (eSCr) (with the omission of the race variable) (Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138; Bellomo et al., Crit Care 2004, 8: R204-12):
Male sex assigned at birth:

- Age 20-24 - eSCr 1.3 mg/dL (115 µmol/l)

- Age 25-39 - eSCr 1.2 mg/dL (106 µmol/l)
- Age 40-65 - eSCr 1.1 mg/dL (97 µmol/l)
- Age > 65 - eSCr 1.0 mg/dL (88 µmol/l)

Female sex assigned at birth:

- Age 20-29 - eSCr 1.0 mg/dL (88 µmol/l)
- Age 30-54 - eSCr 0.9 mg/dL (80 µmol/l)
- Age > 54 - eSCr 0.8 mg/dL (71 µmol/l)

Both the MDRD and the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPl) equations provide a calculation of estimated GFR using serum creatinine, sex assigned at birth, age, and race. Recognition of substantial diversity within population groups has led to multiple organizations no longer including race in the estimated GFR calculations. This approach was investigated by a joint task force of the National Kidney Foundation and the American Society of Nephrology, the result of which was a recommendation to implement the new CKD-EPI creatinine equation which does not use the race variable (Delgado et al., Am J Kidney Dis 2022, 79: 268-88 e1; Inker et al., N Engl J Med 2021, 385: 1737-49).

24:

Patients with acute kidney injury (AKI) and decompensated heart failure (HF) are at risk for poorer outcomes, recurrent admissions, and may quickly advance to chronic kidney disease. Achieving successful diuresis may be challenging and patients with concurrent acute HF and AKI may need advanced therapies (e.g., inotropes). The focus is to achieve optimum decongestion while preserving kidney function (Hollenberg et al., J Am Coll Cardiol 2019, 74: 1966-2011; Bielecka-Dabrowa et al., Curr Heart Fail Rep 2018, 15: 224-38; Takaya et al., Circ J 2015, 79: 1520-5).

25:

Patients hospitalized with acute heart failure (HF) and chronic kidney disease (CKD) are less responsive to loop diuretics and efficacy of established treatments may be reduced because of renal insufficiency. Patients with advanced CKD and acute HF require complicated management to achieve optimal fluid status (Maddox et al., J Am Coll Cardiol 2021, 77: 772-810; Rangaswami et al., Circulation 2019, 139: e840-e78). A study of over 52,000 patients admitted with acute decompensation of chronic systolic or diastolic heart failure concluded that a serum creatinine level of 2.75 mg/dl or greater on admission predicted a high risk for in-hospital mortality (Fonarow et al., JAMA 2005, 293: 572-80).

26:

Multiple studies have identified an elevated level of blood urea nitrogen (BUN) on admission as a marker of poor prognosis for patients presenting with acute decompensated heart failure (HF) (Peterson et al., Circ Cardiovasc Qual Outcomes 2010, 3: 25-32; Rohde et al., J Card Fail 2006, 12: 587-93; Fonarow et al., JAMA 2005, 293: 572-80). A study of over 52,000 patients admitted with acute decompensation of HF concluded that a BUN level of 43 mg/dl or greater on admission was the single best predictor of in-hospital mortality (Fonarow et al., JAMA 2005, 293: 572-80).

27:

Acute or admission hyponatremia, especially sodium < 130 mEq/L(130 mmol/L) in patients with acute heart failure is associated with increased diuretic needs, decreased renal function, prolonged hospitalization, and increased risk of mortality (Ng et al., Cardiology 2014, 128: 333-42).

28:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).
Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with

diverse expertise in varied geographic settings.

29:

A cohort study of over 145,000 heart failure (HF) admissions found that elevated heart rate on presentation in patients with acute decompensated HF is independently associated with a longer length of stay (> 4 days) and higher in-hospital mortality; the finding held true for patients with sinus tachycardia as well as atrial fibrillation (Bui et al., Am Heart J 2013, 165: 567-74.e6). A systematic review identified elevated heart rate as one of the predictors of HF-related readmissions (Choi et al., Heart Lung Circ 2021:).

30:

Lower systolic blood pressure (SBP) on admission may require slower fluid removal with diuresis or ultrafiltration to prevent the development of worsening hypotension. Studies in populations with both systolic and diastolic heart failure (HF) have shown that patients with a SBP below 120 mmHg have worse outcomes. A cohort study of over 48,000 patients with HF with reduced ejection fraction showed that an SPB lower than 120 mmHg on admission was associated with increased in-hospital and post-discharge mortality (Gheorghiade et al., JAMA 2006, 296: 2217-26). A post-hoc analysis of a randomized trial showed that lower SBP on admission, including under 120 mmHg, is an independent predictor of cardiovascular and all-cause mortality in patients with systolic HF (Ambrosy et al., Am Heart J 2013, 165: 216-25). A propensity-score matched observational study in patients with HF with preserved ejection fraction demonstrated that a discharge SBP under 120 mmHg was associated with increased mortality and higher risk of readmission (Tsimploulis et al., JAMA Cardiol 2018, 3: 288-97).

31:

Frailty is a complex condition that occurs because of the aging process and makes individuals more vulnerable to stress, leading to adverse health outcomes like impaired mobility, falls, hospitalizations, and even mortality (Vahedi et al., Int J Geriatr Psychiatry 2022, 37: epub; Kwak and Thompson, Sports Med Health Sci 2021, 3: 1-10; Hoogendijk et al., Lancet 2019, 394: 1365-75). Studies have shown that older individuals identified as frail during hospital admission are at greater risk of in-hospital mortality, longer hospital stays, and functional decline at hospital discharge (Boucher et al., EClinicalMedicine 2023, 59: epub; Cunha et al., Ageing Res Rev 2019, 56: epub). Although there are many validated frailty screening tools available, there has yet to be one that is recognized as the gold standard for identifying frailty. Most of these tools are based on two models: the frailty phenotype and the deficit accumulation model. The frailty phenotype model focuses on the physical aspects of frailty (e.g., strength, walking speed, weight loss), while the deficit accumulation model identifies the number of deficiencies (e.g., functional, cognitive, comorbidities) to calculate a frailty index score (Hoogendijk et al., Lancet 2019, 394: 1365-75). However, both models have limitations in their implementation within the acute care setting, as the frailty phenotype assessment involves performing tests to evaluate the domains (e.g., grip strength, walking speed), and the deficit accumulation model is time-consuming. Therefore, frailty criteria have been developed based on literature and the two models mentioned above to help identify older adults who may have or be at risk for frailty.

32:

The presence of frailty in patients with heart failure is associated with increased hospitalization rates, length of stay, risk for readmission, and mortality (Mollar et al., Am J Cardiol 2022, 183: 48-54; Kwok et al., Int J Cardiol 2020, 300: 184-90; Zhang et al., J Am Med Dir Assoc 2018, 19: 1003-8.e1).

33:

Instruction: This criterion applies to patients who have previously been diagnosed with frailty and are currently experiencing the condition. This information can be retrieved from the medical record, such as the problem list or past medical history. However, describing a patient's appearance as "frail" in a progress note cannot be reviewed in this criterion.

34:

Instruction: The following criteria should be selected based on the patient's baseline or current health status. A patient’s baseline frailty state can affect their risk for severe illness. Similarly, if a patient presents with signs of frailty in the setting of an acute illness or injury, it should also be seen as a marker of illness severity (Theou et al., Clin Geriatr Med 2018, 34: 369-86).

35:

Studies have suggested that incorporating cognitive impairment evaluation with physical characteristics enhances the predictability of frailty (Avila-Funes et al., J Am Geriatr Soc 2009, 57: 453-61). More recently, a systematic review and meta-analysis found that mean Mini-Mental State Examination (MMSE) scores were significantly lower in individuals who were identified as frail by the phenotype model, which focuses on physical function, compared to non-frail and pre-frail individuals. As a result of the strong relationship between physical frailty and cognitive impairment, the authors concluded that cognitive function should be incorporated into frailty assessments (Vahedi et al., Int J Geriatr Psychiatry 2022, 37: epub). Multiple frailty screening tools developed using the cumulative deficit approach assess the cognitive domain, specifically focusing on cognitive impairment or dementia (Mak et al., J Gerontol A Biol Sci Med Sci 2022, 77: 2311-9; Pajewski et al., J Gerontol A Biol Sci Med Sci 2019, 74: 1771-7; Clegg et al., Age Ageing 2016, 45: 353-60).

36:

Multimorbidity refers to the presence of 2 or more chronic conditions or diseases in an individual. Chronic conditions or diseases are those that last for a prolonged period (e.g., 1 year or more) or are permanent and result in disability, decreased quality of life, or require prolonged treatment or rehabilitation (Centers for Disease Control and Prevention, About Chronic Diseases. 2022 [cited Nov 8 2023]; Nicholson et al., J Clin Epidemiol 2019, 105: 142-6; Calderón-Larrañaga et al., J Gerontol A Biol Sci Med Sci 2017, 72: 1417-23). Examples of chronic conditions that have been identified in the literature include but are not limited to (Calderón-Larrañaga et al., J Gerontol A Biol Sci Med Sci 2017, 72: 1417-23):

- Chronic lung disease (e.g., COPD, asthma)
- Chronic kidney disease (e.g., stage 3-4)
- End-stage kidney disease
- Diabetes
- Hypertension
- Congestive heart failure
- Inflammatory bowel disease (e.g., Crohn's disease, ulcerative colitis)
- Mental health disorders (e.g., drug or alcohol dependence, depression, bipolar affective disorder)

37:

According to a recent systematic review and meta-analysis, 72%(0.72) of frail individuals have multimorbidity. The study also found a significant association between frailty and having two or more diseases (Vetrano et al., J Gerontol A Biol Sci Med Sci 2019, 74: 659-66).

38:

According to a study conducted at a Medicare Accountable Care Organization (ACO), the presence of one functional deficit on a frailty index tool in the electronic health record (EHR) was found to be independently associated with a higher risk of hospitalization and injurious falls (Pajewski et al., J Gerontol A Biol Sci Med Sci 2019, 74: 1771-7). When looking specifically at the association between hearing and vision deficits and frailty, a systematic review and meta-analysis found that hearing and vision impairment were associated with 2.5- and 3.0-fold higher odds of frailty, respectively (Tan et al., J Gerontol A Biol Sci Med Sci 2020, 75: 2461-70).

39:

Functional impairment refers to an individual's ability to independently perform activities of daily living (ADLs) (e.g., bathing, dressing, toileting, ambulating), instrumental activities of daily living (IADLs) (e.g., managing medications, finances, preparing meals, usage of the telephone, performing basic housework), or a deficit in hearing or vision that is not corrected by adaptive equipment (e.g., hearing aid, glasses).

40:

Malnutrition has been found to be closely associated with frailty (Lorenzo-López et al., BMC Geriatr 2017, 17: 108). This criterion can be met by one or more of the following:

- Documented weight loss, unintentional, of ≥ 10 pounds or ≥ 5%(0.05) of body weight in the past year (Fried et al., J Gerontol A Biol Sci Med Sci 2001, 56: M146-56)
- Score of ≤ 11 on the Mini Nutritional Assessment-Short Form (MNA-SF) (O'Caoimh et al., Int J Environ Res Public Health 2019, 16: epub; Lorenzo-López et al., BMC Geriatr 2017, 17: 108)

- Body mass index (BMI) < 18.5 kg/m^2 (Mak et al., J Gerontol A Biol Sci Med Sci 2022, 77: 2311-9; Pajewski et al., J Gerontol A Biol Sci Med Sci 2019, 74: 1771-7)
- Documented moderate or severe subcutaneous fat or muscle loss (Cederholm et al., Clin Nutr 2019, 38: 1-9; White et al., JPEN J Parenter Enteral Nutr 2012, 36: 275-83)

41:

Several studies have shown that individuals with polypharmacy are more likely to be frail (Gutiérrez-Valencia et al., Br J Clin Pharmacol 2018, 84: 1432-44). In a recent systematic review and meta-analysis of 66 studies, it was found that 59%(0.59) of older adults with frailty had polypharmacy, defined as the use of five or more medications at the same time. The study also revealed that the prevalence of polypharmacy was highest among older adults with frailty in the hospital setting, with a prevalence of 71%(0.71) (Toh et al., Ageing Res Rev 2023, 83: 101811).

42:

Chronic medications refer to prescription medications that are taken according to a schedule for at least 90 days, as opposed to being taken on an as-needed basis (Masnoon et al., BMC Geriatr 2017, 17: 230). This excludes topical medications, such as creams, or over-the-counter medications.

43:

Patients hospitalized with heart failure, acute myocardial infarction, or pneumonia in addition to a psychiatric comorbidity have higher readmission rates within 30 days than those without a psychiatric diagnosis (Ahmedani et al., Psychiatr Serv 2015, 66: 134-40).

44:

The term "severe and persistent mental illness, cognitive impairment" also called "serious and persistent mental illness," refers to a mental illness or cognitive impairment that is characterized by severe symptoms of prolonged duration that require long-term treatment. The illness causes impaired functioning that interferes significantly with primary activities of daily living (ADLs) and results in an inability to maintain independent functioning without treatment, support, and rehabilitation for a prolonged or indefinite period of time. Examples include, but are not limited to, schizophrenia and other psychotic disorders, bipolar disorder, dementia, Alzheimer's disease, and severe cases of major depression.

45:

Patient unreliable refers to the following:

- Inability or unwillingness to follow an outpatient monitoring or treatment plan
- Inability to understand the signs of deterioration and recognize the need for prompt medical attention

46:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

47:

High flow nasal cannula (HFNC) delivers fraction of inspired oxygen (FiO_2) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the FiO_2. The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (face mask or nasal cannula with a flow rate of 15 L/min); however, did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

48:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).
The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.
Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

49:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO_2) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO_2 with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO_2 retention, where a precise inspired FiO_2 is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO_2. The following are estimates of O_2 delivered at the associated flow rates:
LOW-FLOW SYSTEMS
Nasal cannula
Room air 21%(*0.21*)
1L 24%(*0.24*) **4L** 36%(*0.36*)
2L 28%(*0.28*) **5L** 40%(*0.40*)
3L 32%(*0.32*) **6L** 44%(*0.44*)
Simple oxygen masks

- 35-50%(*0.35-0.50*) FiO_2 (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO_2 in the mask

Venturi masks

- 24-50%(*0.24-0.50*) FiO_2 (4-12 L/min)

Partial rebreathing masks

- 40-70%(*0.40-0.70*) FiO_2 (6-10 L/min)

Non-rebreathing masks

- 60-80%(*0.60-0.80*) FiO_2 (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS
Air-entrainment masks/nebulizers

- 24-40%(0.24-0.40) FiO_2

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO_2

50:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

51:

Intravenous (IV) loop diuretics (e.g., furosemide, torsemide, bumetanide, ethacrynic acid) are administered to relieve the symptoms of dyspnea and congestion without excessively reducing intravascular volume. Due to their relatively short half-life, diuretic effectiveness can be enhanced by continuous administration or multiple boluses daily. Careful monitoring of daily weights, orthostatic vital signs, intake and output, electrolytes, and renal function are key components of diuretic therapy. If a patient does not initially respond to IV diuretics, other options may be considered, including increasing the dose to ensure adequate drug levels reach the kidneys and adding a second diuretic, typically a thiazide, to the loop diuretic in order to improve diuretic responsiveness (Heidenreich et al., Circulation 2022, 145: e895-e1032; Maddox et al., J Am Coll Cardiol 2021, 77: 772-810; Rosendorff et al., Circulation 2015).

52:

Extracorporeal membrane oxygenation (ECMO) or extracorporeal life support (ECLS) provides oxygen and removes carbon dioxide from blood. ECMO/ECLS facilitates the recovery of cardiopulmonary function when other measures fail to improve gas exchange. Indications for the use of ECMO/ECLS in children include refractory septic or cardiogenic shock, cardiac arrest as a bridge to transplant, or refractory hypoxemic or hypercarbic respiratory failure. There is clear evidence to support the use of ECMO/ECLS in the pediatric population, but definitive studies in adults are lacking. However, in adults, observational studies and other data support the use of ECMO/ECLS for severe refractory, hypoxemic, or hypercarbic respiratory failure (Mehta et al., American College of Cardiology Perspectives and Analysis 2018; Napp et al., Herz 2017, 42: 27-44; Guirand et al., J Trauma Acute Care Surg 2014, 76: 1275-81). ECMO/ECLS incorporates the use of an external circuit, a cannula for vascular access, a pump to propel the blood through the circuit, and an artificial membrane that oxygenates the blood. Common complications include hemorrhage, hemolysis, and infection. Stroke, embolism, and seizures in children have also been reported. Infants with intraventricular hemorrhage (IVH) or those at high risk for IVH (i.e., weight less than 2000 grams, age less than 35 weeks) are not candidates for this therapy (Abrams et al., J Am Coll Cardiol 2014, 63: 2769-78; Domico et al., Pediatr Crit Care Med 2012, 13: 16-21; MacLaren et al., Intensive Care Med 2012, 38: 210-20; Peek et al., Health Technol Assess 2010, 14: 1-46; Mugford et al., Cochrane Database Syst Rev 2008: CD001340).

53:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

54:

An intra-aortic balloon pump (IABP) is a mechanical circulatory support device that utilizes a catheter-mounted intravascular device positioned in the descending thoracic aorta. The balloon automatically inflates during early diastole, increasing coronary blood flow and deflates during early systole, reducing afterload as the left ventricle ejects. Indications for use include cardiogenic shock, temporary support of left ventricular function after myocardial infarction (MI), preservation of myocardial viability, or refractory ventricular arrhythmias post-MI or preoperative bridge before coronary artery bypass graft (CABG) surgery. Contraindications include aortic regurgitation and aortic dissection.

55:

A percutaneous left ventricular assist device (VAD) is used to protect the heart during invasive procedures or temporarily for short-term mechanical circulatory support while treatment decisions are made.

56:

In some facilities a ventricular assist device (VAD) will be used as a bridge to heart or heart-lung transplant. This may stabilize a patient long enough to survive until transplant, but the patient is still fragile and relatively unstable. **Instruction**: In this clinical setting, the patient may require Critical care until a donor organ is available. Due to the restrictions associated with these devices, the Critical level of care is required for percutaneous VADs.

57:

Management of patients admitted with a heart failure (HF) exacerbation may include supplemental oxygen, diuretic, guideline-directed medical therapy (GDMT), which includes angiotensin-converting enzyme inhibitor (ACEi) or angiotensin receptor blocker (ARB) or angiotensin receptor neprilysin inhibitor (ARNi), beta-blocker, mineralocorticoid receptor antagonists (MRA), and sodium-glucose cotransporter 2 inhibitors (SGLT2i), cardiac monitoring, and deep vein thrombosis (DVT) prophylaxis (Maddox et al., J Am Coll Cardiol 2021, 77: 772-810).

58:

The American College of Cardiology and American Heart Association (ACC/AHA) recommend the following labs be monitored during the use of intravenous (IV) diuretics or active titration of heart failure (HF) medications (Heidenreich et al., Circulation 2022, 145: e895-e1032):

- Daily serum electrolytes
- Blood urea nitrogen (BUN)
- Creatinine

59:

Instruction: For this criteria point, medication adjustment refers to non-diuretic medications used in the treatment of acute heart failure (e.g., vasodilators, afterload reducers, antihypertensives).

60:

Instruction: This line of criteria applies when there is a change in diuretic medications, including:

- Administering a higher dose compared to the dose used in the Emergency Department (ED)
- Changing of the diuretic being used
- Adding a second diuretic agent

61:

The CMS Two-Midnight Rule generally requires that the medical record support a patient’s need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.
For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.
NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.
(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

62:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

63:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

64:

Patients with heart failure exacerbation who are slow to respond to diuretic therapy may benefit from evaluation by cardiologist or heart failure specialist and may require further assessment or additional treatments, such as (Hollenberg et al., J Am Coll Cardiol 2019, 74: 1966-2011):

- Higher dose of diuretics or addition of a second diuretic
- Invasive hemodynamic monitoring through right heart catheterization or pulmonary artery catheter
- Re-evaluation of comorbidities and/or alternate diagnosis such as ischemic heart disease, lung disease, or thyroid disease

65:

Resolution of heart failure signs and symptoms is desired by the time of discharge as higher rates of mortality and readmission are seen in patients discharged with persistent congestion. Patients with dyspnea, orthopnea, jugular venous distention (JVD), and persistent rales may need further diuresis. Although peripheral edema (e.g., edema of lower extremities) can a sign of congestion, eliminating peripheral edema may not be always feasible due to other coexisting conditions that can contribute to peripheral edema, such as chronic venous insufficiency, lymphedema, or hypoalbuminemia (Heidenreich et al., Circulation 2022, 145: e895-e1032; Hollenberg et al., J Am Coll Cardiol 2019, 74: 1966-2011).

66:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

67:

Instruction: Application of this criteria point applies to acute deconditioning which has occurred during the current hospitalization.

68:

Hypertension is defined as a systolic blood pressure (SBP) of greater than 130 mmHg or a diastolic blood pressure (DBP) of 80 mmHg or greater. Recommendations for initial management may include a conservative approach with diet and lifestyle modification with or without medications. The goal of antihypertensive therapy is to reduce morbidity and mortality. In non-urgent situations, blood pressure (BP) control is achieved in the outpatient setting. Goals for target BP level should be individualized, but a SBP less than 130 mmHg and a DBP less than 80 mmHg are desirable. Populations at high risk include patients with the following conditions (Whelton et al., Hypertension 2018, 71: 1269-324):

- Clinical cardiovascular disease or 10-year atherosclerotic cardiovascular disease (ASCVD) risk 10%(0.1)
- Heart failure
- Stable ischemic heart disease
- Chronic kidney disease
- Chronic kidney disease after renal transplantation

69:

When a patient with hypokalemia is not responding to potassium repletion, the magnesium level should be checked since hypomagnesemia could be contributing to the refractory hypokalemia.

70:

In patients with no underlying heart disease, occasional, nonconsecutive premature ventricular contractions (PVCs) less than 6/min are common and not considered a concerning finding. Three or more consecutive PVCs at a rate of 100 bpm or more that terminate spontaneously within 30 seconds are considered nonsustained ventricular tachycardia (Marine, Card Electrophysiol Clin 2016, 8: 525-43; Buxton et al., Circulation 2006, 114: 2534-70).

71:

Following an initial 12-lead electrocardiogram (ECG) demonstrating no abnormal findings in the setting of hyperkalemia or hypokalemia, continued monitoring may be accomplished by telemetry (Bridgeman et al., Nephrol Dial Transplant 2019, 34: iii45-iii50).

72:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

73:

Resolution of heart failure signs and symptoms is desired by the time of discharge as higher rates of mortality and readmission are seen in patients discharged with persistent congestion. Patients with dyspnea, orthopnea, jugular venous distention (JVD), and persistent rales may need further diuresis. Pre-sacral, scrotal, and peroneal edema may be signs of marked congestion. Although peripheral edema (e.g., edema of lower extremities) can be a sign of congestion, eliminating peripheral edema may not be always feasible due to other coexisting conditions that can contribute, such as chronic venous insufficiency, lymphedema, or hypoalbuminemia (Heidenreich et al., Circulation 2022, 145: e895-e1032; Hollenberg et al., J Am Coll Cardiol 2019, 74: 1966-2011).

74:

Pre-sacral, scrotal, and peroneal edema may be signs of marked congestion. Although peripheral edema (e.g., edema of lower extremities) can be a sign of congestion, eliminating peripheral edema may not be always feasible due to other coexisting conditions that can contribute to peripheral edema, such as chronic venous insufficiency, lymphedema, or hypoalbuminemia (Hollenberg et al., J Am Coll Cardiol 2019, 74: 1966-2011).

75:

Some patients may require invasive evaluation to guide management. Patients who are persistently symptomatic despite treatment, patients whose fluid status, perfusion, or systemic or pulmonary vascular resistance is uncertain, or patients whose renal function worsens with heart failure (HF) treatment may benefit from right heart catheterization. Coronary artery disease is a leading cause of HF, and left heart catheterization may be needed if coronary artery disease is suspected (Heidenreich et al., Circulation 2022, 145: e895-e1032).

76:

Instruction: Infrequent titration of low dose inotropes can be provided at the Acute level of care. A less intensive level of care may be appropriate for patients requiring low dose inotropes on a long-term basis.

77:

Inotropes (e.g., dopamine, dobutamine, and milrinone) can be used in patients with severe systolic dysfunction resulting in hypotension and low cardiac output. These medications can be effective in treating patients who are unable to tolerate or do not respond to vasodilators and diuretics. The goal of treatment is to maintain systemic perfusion and improve end-organ hypoperfusion. Temporary use of inotropes is recommended for hospitalized patients in cardiogenic shock, and more prolonged use as a bridge therapy may be indicated in advanced heart failure (HF) patients (stage D) who are awaiting mechanical circulatory support (MCS) or heart transplant. They may also be indicated as a palliative therapy for those unable to undergo MCS or transplantation. Patients on inotropes require close monitoring as they are either unstable or have advanced disease, and inotropes themselves carry risks such as arrhythmias (Heidenreich et al., Circulation 2022, 145: e895-e1032; Maddox et al., J Am Coll Cardiol 2021, 77: 772-810).

78:

Guidelines recommend preprandial glucose targets of 80-130 mg/dL (4.4-7.2 mmol/L) and a random blood glucose target range of 100-180 mg/dL (5.6-10.0 mmol/L) in hospitalized patients. In stable patients with previous tight glycemic control, more stringent targets may be appropriate. In patients with severe comorbidities, a history of hypoglycemia, terminal illness, late onset diabetes, and in older patients, less stringent targets may be appropriate. The preferred method for achieving and maintaining glucose control in non-critically ill patients is scheduled subcutaneous insulin administration. Guidelines discourage the sole use of sliding scale insulin in the inpatient hospital setting. Rather, an insulin regimen consisting of basal, prandial, and correctional insulin is recommended in hospitalized patients (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

79:

Instruction: Insulin adjustments should be based on blood glucose values obtained by lab draw or glucose monitor. Intermittent insulin must be administered at least 3 times in 24 hours. If the patient is on a continuous insulin infusion, the rate must be adjusted at least twice in 24 hours. A dose that is held based on a normal or low blood glucose value should be included in the count of doses administered within the 24-hour period.

80:

A severe chronic obstructive pulmonary disease (COPD) exacerbation may be indicated by the following signs (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; Celli et al., Eur Respir J 2004; 23(6): 932-946):

- Accessory respiratory muscle use
- Central cyanosis, new onset or worsening
- Paradoxical chest wall movements

81:

Venous blood gases (VBG) may be used to assess a patient's ventilation or metabolic status. Venous pH or HCO_3 values closely approximate corresponding values derived from arterial blood gases (ABG) in conditions such as chronic obstructive pulmonary disease (COPD), respiratory distress syndrome, neonatal sepsis, renal failure, pneumonia, diabetic ketoacidosis, and status epilepticus. VBG should not be used to determine oxygenation. Patients with chronic CO_2 retention hospitalized for acute respiratory illness will exhibit increased PCO_2 levels. PCO_2 values should be compared with baseline values when available and other signs and symptoms of respiratory distress should be considered when determining the appropriate level of care (e.g., worsening dyspnea, high respiratory rate, decreased oxygen saturation, confusion, drowsiness) (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; McKeever et al., Thorax 2016, 71: 210-5).

82:

In the setting of dyspnea, patients prefer sitting because they are unable to breathe effectively in the recumbent position.

83:

Instruction: These criteria apply to the administration of a bronchodilator via a nebulizer, an inhaler, or an intravenous route. According to the available evidence, inhalers produce outcomes that are at least equivalent to nebulizers. Further research is needed to determine the optimal inhalation device according to individual patient characteristics and disease state. Education on correct usage is vitally important to ensure effective medication delivery to the lungs (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; van Geffen et al., Cochrane Database Syst Rev 2016: CD011826).

84:

Evidence in the medical literature to support this criterion or the effectiveness of this intervention or service is mixed or unclear. The criteria point reflects current best evidence and practice. It is the product of a peer review process involving multiple clinicians with diverse expertise in varied practice and geographic settings.

85:

The administration of systemic corticosteroids in patients with a chronic obstructive pulmonary disease (COPD) exacerbation improves lung function, decreases hypoxemia, shortens recovery time, reduces hospital length of stay, and prevents early reoccurrence. Evidence demonstrates daily administration of 40 milligrams of prednisone in patients who have been hospitalized for more than 5 days is not inferior to a 14-day course in terms of early relapse. This shortened regimen is supported by current guidelines. Studies have shown low-dose oral steroids demonstrate equivalent outcomes to intravenous therapy in the treatment of COPD. Nebulized budesonide may be an alternative treatment for exacerbations with similar benefits to intravenous methylprednisolone in select patients, according to the GOLD guidelines (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024).

86:

Hyponatremia (Seay et al., Am J Kidney Dis 2020, 75: 272-86)

87:

Hypovolemic hyponatremia may be treated with sodium chloride 0.9% (0.009). For severe hyponatremia, treatment with an infusion of sodium chloride 3% (0.03) may be required. Rapid correction of serum sodium can cause spastic quadriparesis or quadriplegia, pseudobulbar palsy, dysarthria, mutism, pupillary or oculomotor abnormalities, stupor, coma, ataxia, dystonia, osmotic demyelinating syndrome, or even death (Weismann et al., Endocr Connect 2016:).

88:

Instruction: This criterion includes aspiration pneumonitis. Aspiration is marked by the abnormal movement of oropharyngeal or gastric contents into the lower respiratory tract. Aspiration pneumonitis occurs when acidic gastric contents cause inflammation of the lung parenchyma. In the early stages, aspiration pneumonitis is not considered an infectious process as gastric acid prevents microbial growth. Anti-infective therapy should be considered if symptoms (e.g., shortness of breath, wheezing, hypoxemia) fail to resolve within 48 hours (Neill and Dean, Curr Opin Infect Dis 2019, 32: 152-7).

89:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

90:

Mental status change may include confusion, disorientation, delirium, or increasing lethargy (Lacey et al., Ann Med 2019, 51: 232-51).
Instruction: Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma, stupor, or obtundation (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

91:

Expected treatments for patients with bacterial pneumonia include anti-infective and oximetry or blood gas monitoring (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67; Cohoon et al., Am J Med 2018, 131: 307-16 e2; Kaysin and Viera, Am Fam Physician 2016, 94: 698-706; Jain et al., N Engl J Med 2015:[e-pub]).

92:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

93:

Surgical therapies or percutaneous interventions that are commonly considered, in the management of heart failure (HF) include (Yancy et al., J Card Fail 2017, 23: 628-51):

- Coronary revascularization (e.g., Coronary artery bypass graft (CABG), angioplasty, stenting)
- Aortic valve replacement
- Mitral valve replacement or repair
- Septal myectomy or alcohol septal ablation for hypertrophic cardiomyopathy
- Surgical ablation for ventricular arrhythmia
- Mechanical circulatory support (MCS)
- Cardiac transplantation

The most important considerations in the decision to proceed with a surgical or interventional approach include coronary anatomy that is amenable to revascularization and the appropriate associated guideline-directed medical therapy (GDMT).

94:

Reference: (Jordan and Caesar, BMJ Qual Improv Rep 2015, 4:)

95:

The "R on T phenomenon" is the superimposition of an ectopic beat on the T wave of a preceding beat. Early observations suggest that R on T is likely to initiate sustained ventricular tachyarrhythmias.

96:

When T and U waves fuse, they form a TU fusion wave on an electrocardiogram. Severe hypokalemia may be associated with a large U wave created by T and U wave fusion (Zieg et al., Acta Paediatr 2016, 105: 762-72; Levis, Perm J 2012, 16: 53).

97:

The U wave occasionally follows the T wave on an electrocardiogram. A prominent upright U wave is associated with hypokalemia. When the U wave exceeds the T wave amplitude, the serum potassium level is less than 3 mEq/L. Severe hypokalemia may be associated with a giant U wave created by T and U wave fusion (Sandau et al., Circulation 2017, 136: e273-e344; Zieg et al., Acta Paediatr 2016, 105: 762-72).

98:

Symptoms of hyponatremia may include nausea, lethargy, headache, and muscle weakness. If hyponatremia worsens, symptoms may progress to mental status changes (Glasgow Coma Scale (GCS) 9-14). Patients with acute coma, stupor, obtundation, and seizures should be reviewed at a higher level of care due to the need for more frequent neurological monitoring (Weismann et al., Endocr Connect 2016:).

99:

Symptoms of hyponatremia include:

- Coma
- Headache
- Lethargy
- Nausea
- Obtundation
- Seizures
- Stupor
- Vomiting

100:

Fluid restriction should be avoided during the first 24 hours of therapy with conivaptan or tolvaptan as it may increase the likelihood of an overly rapid correction of serum sodium and may lead to the development of cerebral edema.

101:

The CMS Two-Midnight Rule generally requires that the medical record support a patient’s need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.
For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.
Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status;** those who do not meet criteria would be appropriate for secondary review.
(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

102:

Expected progress after 48-72h

- O_2 saturation at or close to baseline
- Renal function stabilized

- Activities at or close to baseline
- Preparing for discharge within the next 24 hours.

If not, consider (Heidenreich et al., Circulation 2022, 145: e895-e1032; Yancy et al., J Card Fail 2017, 23: 628-51):

- Optimizing heart failure(HF) medication regimen to ensure relief from congestive symptoms
- Echocardiogram for patients with a significant clinical change or a potential cardiac function change
- Physical therapy consult for ambulation and prevention of deconditioning
- Cardiology or nephrology consultation if not previously obtained
- Evaluation for Ventricular Assist Device (VAD) or transplant

103:

Functional impairment refers to any condition that results in activity limitations impacting the patient's ability to ambulate, transfer, and/or perform activities of daily living (ADLs) or instrumental activities of daily living (IADLs). In pediatric patients, functional impairment includes limitations that impact the patient's developmental progress, such as deficits in mental status, sensory functioning, communication, motor functioning, feeding, and respiratory status. The patient will require assistance with one or more limitations.

104:

Functional impairments may be safely managed in the home environment when the patient is willing and able to perform self-care independently with or without assistive devices, has sufficient caregiver support available to adequately meet the patient's needs or home care services are arranged.

105:

Application of this criteria point is appropriate for a sustained finding based on room air readings unless the criteria state otherwise. If a patient remains hypoxic while on oxygen, room air assessment is not required. Oxygen saturation (O2 sat) is considered sustained finding when it is abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes.

106:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

107:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

108:

Guidelines recommend multidisciplinary heart failure (HF) disease management programs to reinforce the education of guideline-directed medical therapy (GDMT), address the different barriers to change, and strive to minimize the risk of readmission for HF (Heidenreich et al., Circulation 2022, 145: e895-e1032; Yancy et al., J Card Fail 2017, 23: 628-51; Feltner et al., Ann Intern Med 2014, 160: 774-84).
In order to ensure that patients are clinically appropriate for discharge and to decrease the risk of readmission, the medical practitioner should perform a thorough assessment of overall prognosis and functional status. Patients with HF are managed with a core group of medications that may include diuretics, angiotensin converting enzyme inhibitors (ACEi) or angiotensin receptor blockers (ARB) or angiotensin receptor neprilysin inhibitors (ARNi), beta blockers, mineralocorticoid receptor antagonists (MRA), and sodium-glucose cotransporter 2 inhibitors (SGLT2i) (Heidenreich et al., Circulation 2022, 145: e895-e1032). Prior to discharge, these medications should be adjusted and maximized to the most effective dose when appropriate. Patients should be encouraged to participate in smoking cessation programs and regular exercise programs. All patients should be encouraged to monitor their sodium intake and daily weight. Alcohol, NSAID, and illicit drug use should be discouraged. Written discharge and teaching

material should be provided to the patient or family at the time of discharge (Yancy et al., J Card Fail 2017, 23: 628-51; Feltner et al., Ann Intern Med 2014, 160: 774-84).

109:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

110:

Providing education and encouraging active involvement in the treatment plan is an important part of the discharge process for heart failure (HF) patients. Discharge education helps patients to understand and learn HF symptoms and treatment choices, current medication list (i.e., indication, necessity, and possible side effects of HF medications), the importance of self-care and monitoring (e.g., exercising, liquid restriction if indicated, monitoring salt intake, smoking cessation, avoiding excessive alcohol) (McDonagh et al., Eur Heart J 2021, 42: 3599-726). Failure to comprehend discharge teaching or to adhere to the treatment plan can lead to hospital readmission as seen in examples below:

- Increases in body weight and worsening of symptoms are often the result of nonadherence with diet and medication. Patients with a daily weight gain of greater than 2 pounds are at risk for developing symptoms that require emergency treatment or hospitalization. Enrollment in a weight management program may increase adherence to weight monitoring, improvement in New York Heart Association classification, and a reduction in readmissions (Wang et al., J Cardiovasc Nurs 2014, 29: 528-34).
- Physical deconditioning related to inactivity may contribute to exercise intolerance and adversely affect clinical status. Physical activity can improve exercise capacity, reduce symptoms of depression, and improve quality of sleep. It should be encouraged in patients with HF except during periods of acute exacerbations or in patients with myocarditis (Leifer et al., Med Sci Sports Exerc 2014, 46: 219-24; Tu et al., Eur J Heart Fail 2014, 16: 749-57; Suna et al., Eur J Cardiovasc Nurs 2014:). Exercise-based cardiac rehabilitation can also improve exercise capacity, reduce hospital admissions, and improve health-related quality of life. It may reduce long-term mortality (Taylor et al., Cochrane Database Syst Rev 2014, 4: CD003331; Lewinter et al., Eur J Prev Cardiol 2014:).

- NSAIDs can cause sodium retention and peripheral vasoconstriction as well as increase the toxicity of diuretics and angiotensin-converting enzyme inhibitors.
- Avoidance of tobacco is encouraged because of the vasoconstrictor and proinflammatory activity of nicotine.
- Guidelines recommend limiting alcohol intake because it is toxic to the liver, is detrimental to other organs, decreases the motivation to adhere to diet, and is an addictive substance.

(Yancy et al., Circulation 2013, 128(16):e240-319; Riegel et al., Circulation 2009; 120(12): 1141-1163; Chaudhry et al., Circulation 2007; 116(14): 1549-1554)

111:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear

information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

112:

Guidelines advocate scheduling an early follow up visit within 7 to 14 days or a telephone follow up within 3 days of hospital discharge for heart failure to reinforce the care plan and minimize the risk for readmission (Yancy et al., J Card Fail 2017, 23: 628-51; Albert et al., Circ Heart Fail 2015, 8: 384-409).

113:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

114:

Outpatient management contraindicated refers to the following:
- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

115:

Outpatient management unavailable refers to the following:
- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

116:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:
- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management

- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

117:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

118:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.
Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.
Evaluation and Treatment
Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

119:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

120:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies

- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

121:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

122:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

123:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

124:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.
The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

125:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility.

Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

126:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

127:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

128:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

129:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.